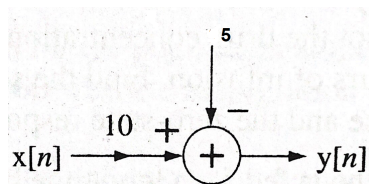
EED-201 (Signals and Systems): TUTORIAL 3

1. Determine whether the system is linear, time invariant, causal, memoryless, BIBO stable, invertible:
   i) $y(t) = u(x(t))$ ii) $y(t) = x(t/2)$ iii) $y[n] = |x[n+1]|$ iv) $y[n] = nx[n]$
   v) $y[n] = e^{x[n]}$ vi) $y(t) = x(sin(t))$ vii) $y[n] = x[-n]$ viii) $y(t) = sin(x(t))$.

2. Consider following continuous time systems and determine whether the systems are linear and time invariant:
   i) $y(t) = \mathbf{T}\{x(t)\} = \sum_{k=-\infty}^{\infty} x(t)\delta(t - kT_s)$ ii) $y(t) = \mathbf{T}\{x(t)\} = \frac{1}{T}\int_{t-T/2}^{t+T/2} x(\tau)d\tau$.

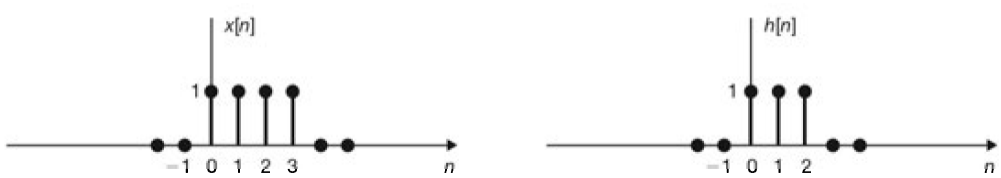
3. Consider a discrete-time system represented by an operator T such that:
   $y[n] = \mathbf{T}\{x[n]\} = x^*[n]$;
   where $x^*[n]$ is the complex conjugate of $x[n]$. Determine whether the system is linear if the input is scaled by an arbitrary complex valued constant.

4. Show that the system below is non-linear, time invariant, static and invertible.

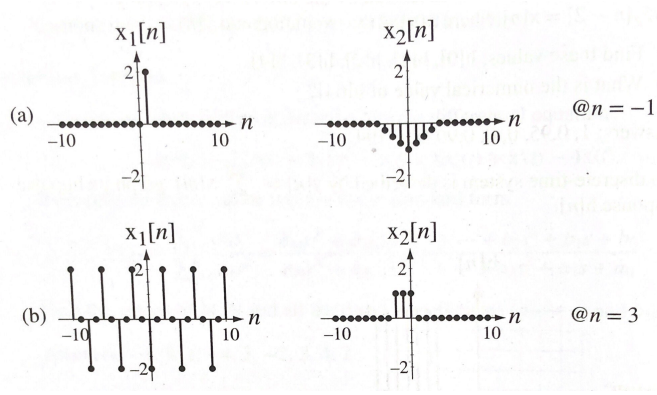


5. Show that a system described by $y(t) = \frac{dx(t)}{dt}$ is not invertible and a system described by $y(t) = \int_{-\infty}^{t} x(\lambda)d(\lambda)$ is invertible.

6. Evaluate $y[n] = x[n] * h[n]$, where $x[n]$ and $h[n]$ are shown below, (a) by an analytical technique, and (b) by a graphical method.



7. For each pair of signals $x_1[n]$ and $x_2[n]$, find the numerical value of $y[n] = x_1[n] \circledast x_2[n]$ at the indicated values of n.



8. Compute the convolution of:
   i) $x[n] = \alpha^n u[n], h[n] = \beta^n u[n]$ ii) $x[n] = \alpha^n u[n], h[n] = \alpha^{-n} u[-n], 0 < \alpha < 1$.